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Inventor(s)

Cakulev; Violeta et al.

Systems and methods for quality of service monitoring activated by policy control function

Abstract

A device may include a processor configured to determine that Quality of Service (QoS) monitoring is to be performed for a communication session; generate a QoS monitoring policy for the communication session; and provide the QoS monitoring policy to a Session Management Function (SMF) associated with the communication session. The processor may be further configured to receive a QoS monitoring report for the communication session from the SMF and perform an action based on the received QoS monitoring report.

Inventors: Cakulev; Violeta (Millburn, NJ), Yan; Lixia (Basking Ridge, NJ), Patil; Sudhakar Reddy (Flower Mound, TX), Reynolds; Anthony Clay (Rhome, TX)

Applicant: Verizon Patent and Licensing Inc. (Basking Ridge, NJ)

Family ID: 89432888

Assignee: Verizon Patent and Licensing Inc. (Basking Ridge, NJ)

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Background/Summary

BACKGROUND INFORMATION

(1) To satisfy the needs and demands of users of mobile communication devices, providers of wireless communication services continue to improve and expand available services and networks used to deliver such services. One aspect of such improvements includes the development of wireless access networks and options to utilize such wireless access networks. For example, a wireless access network may need to manage a large number of communication sessions. Maintaining a large number of communication sessions may pose various challenges.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an environment according to an implementation described herein;
- (2) FIG. 2 illustrates exemplary components of a Fifth Generation (5G) core network according to an implementation described herein;
- (3) FIG. 3 is a diagram illustrating exemplary components of a device that may be included in a component of an environment according to an implementation described herein;
- (4) FIG. 4 is a diagram illustrating exemplary components of a Policy Control Function (PCF) device according to an implementation described herein;
- (5) FIG. 5A is a diagram illustrating exemplary components of a Quality of Service (QoS) monitoring criteria database (DB) according to an implementation described herein;
- (6) FIG. 5B is a diagram illustrating exemplary components of a QoS policy rules DB according to an implementation described herein;
- (7) FIG. 6 illustrates a flowchart for initiating QoS monitoring according to an implementation described herein;
- (8) FIG. 7 illustrates a flowchart for batch reporting of QoS monitoring reports according to an implementation described herein;
- (9) FIG. 8 illustrates a first exemplary signal flow according to an implementation described herein; and
- (10) FIG. 9 illustrates a second exemplary signal flow according to an implementation described herein.

DETAILED DESCRIPTION OF EMBODIMENTS

- (11) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements.
- (12) Wireless network, such as Fourth Generation (4G) wireless networks, Fifth Generation (5G) wireless networks, and/or other types of wireless networks, continue to add new users and manage an increasing number of communication sessions. A communication session, associated with a user equipment (UE) device attached to a 4G or 5G core network via a Radio Access Network (RAN), may provide Internet Protocol (IP) and/or other kind of connectivity between the UE device and an application server in a network, another UE device, and/or another type of device.
- (13) A 5G core network may implement different types of services, such as, for example, an enhanced Mobile Broadband (eMBB) service for Voice over Internet Protocol (VoIP) telephone calls and/or data sessions for accessing Internet websites; a massive Internet of Things (IoT) service for IoT devices; an Ultra-Reliable Low Latency Communication (URLLC) service for devices such as medical monitoring devices, autonomous vehicles, industrial automation, etc.; and/or other types of wireless communication services.
- (14) Different services may have different Quality of Service (QoS) requirements. A core network may define QoS monitoring to measure one or more QoS parameters for a communication session in real-time. For example, a Policy Control Function (PCF) may activate a QoS monitoring policy on a data flow for a communication session, in response to a request from an Application Function (AF) or a Network Exposure Function (NEF). The PCF may provide the QoS monitoring policy to a Session Management Function (SMF). In response, the SMF may send a QoS monitoring request to a User Plane Function (UPF) associated with the communication session. The UPF may obtain data for QoS parameters and report the obtained data to the SMF based on a frequency specified by QoS monitoring request. The SMF may then in turn report the QoS monitoring results to the PCF based on a reporting frequency specified by the QoS monitoring policy and/or a configuration associated with the PCF.
- (15) Reporting QoS monitoring data may result in a significant number of signaling messages being sent in the network. In situations where immediate action in response to a QoS monitoring report is not needed, a mechanism to reduce signaling in the network may improve the efficiency of

the network. In addition, a PCF may need to react to information included in a QoS monitoring report. For example, the PCF may need to adjust a QoS class for a communication session in response to information included in a received QoS monitoring report. In such situations, a mechanism to initiate QoS monitoring, which is independent of QoS monitoring triggered by requests from an AF or an NEF, may be needed for efficient functioning of the core network.

(16) Implementations described herein relate to systems and methods for QoS monitoring triggered by a PCF and batch reporting of QoS monitoring reports. For example, a PCF in a 5G core network may be configured to determine that QoS monitoring is to be performed for a communication session, generate a QoS monitoring policy for the communication session in response, and provide the QoS monitoring policy to an SMF associated with the communication session. The QoS monitoring policy may instruct the SMF to measure values for at least one of an uplink delay (also referred to as latency), a downlink delay, a round-trip delay, a packet loss rate, a jitter rate, a throughput rate, and/or another QoS parameter associated with the communication session. The PCF may be further configured to receive a QoS monitoring report for the communication session from the SMF and perform an action based on the received QoS monitoring report.

(17) The action may include sending the received QoS monitoring report to an AF, NEF, and/or other types of NFs. Additionally, or alternatively, the action may include applying a policy to the SMF based on the received QoS monitoring report, such as, for example, changing a QoS class for the communication session, instructing the SMF to collect additional QoS information relating to the communication session, instructing the SMF to transfer the communication session to a particular UPF, assigning the communication session to a particular network slice, and/or applying another type of policy to the communication session.

(18) Determining that QoS monitoring is to be performed for the communication session may include at least one of: receiving, from a Charging Function (CHF), a notification indicating that QoS monitoring is to be performed for the communication session; receiving a subscriber profile update indicating that QoS monitoring is to be activated for communication sessions associated with a subscriber; determining that the communication session satisfies a trigger condition for activating QoS monitoring; determining that QoS monitoring is to be established for communication sessions associated with a particular Data Network Name (DNN) or Access Point Name (APN); determining that QoS monitoring is to be established for every communication session; and/or another type of determination.

(19) Furthermore, the PCF may be configured to collect QoS monitoring reports from SMF s and provide the collected QoS monitoring reports to an AF and/or NEF in a batch report. For example, the PCF may be configured to generate a batch reporting rule that includes a reporting trigger condition, detect that the reporting trigger condition is satisfied, and send the batch report to the AF or NEF (and/or other types of NFs in core network **140**), in response to detecting that the reporting trigger condition is satisfied. Detecting the reporting trigger condition may include at least one of detecting that a size of data included in the plurality of QoS monitoring reports has reached a data size threshold, detecting that a particular time period has elapsed, detecting that a particular number of QoS monitoring reports have been received for a particular subscriber, and/or detecting another type of reporting trigger condition has been satisfied.

(20) FIG. 1 is a diagram of an exemplary environment **100** in which the systems and/or methods described herein may be implemented. As shown in FIG. 1, environment **100** may include UE devices **110-A** to **110-N** (referred to herein collectively as “UE devices **110**” and individually as “UE device **110**”), base stations **120-A** to **120-M** (referred to herein collectively as “base stations **120**” and individually as “base station **120**”) in RAN **130**, core network **140**, and packet data networks (PDNs) **150-A** to **150-Y** (referred to herein collectively as “PDNs **150**” and individually as “PDN **150**”).

(21) UE device **110** may include any device with cellular wireless communication functionality. For example, UE device **110** may include a handheld wireless communication device (e.g., a

mobile phone, a smart phone, a tablet device, etc.); a wearable computer device (e.g., a head-mounted display computer device, a head-mounted camera device, a wristwatch computer device, etc.); a laptop computer, a tablet computer, or another type of portable computer; a desktop computer; a customer premises equipment (CPE) device, such as a set-top box or a digital media player (e.g., Apple TV™, Google Chromecast™, Amazon Fire TV™, etc.), a WiFi access point, a smart television, etc.; a portable gaming system; a global positioning system (GPS) device; a home appliance device; a home monitoring device; and/or any other type of computer device with wireless communication capabilities and a user interface. In some implementations, UE device **110** may communicate using machine-to-machine (M2M) communication, such as Machine Type Communication (MTC), and/or another type of M2M communication for IoT applications.

(22) RAN **130** may include base stations **120**. Base station **120** may enable UE device **110** to communicate with core network **140**. Base station **120** may be configured for one or more Radio Access Technology (RAT) types. For example, base station **120** may include a 5G New Radio (NR) base station (e.g., a gNodeB) and/or a 4G Long Term Evolution (LTE) base station (e.g., an eNodeB). Each base station **120** may include devices and/or components to enable cellular wireless communication with UE devices **110**. For example, base station **120** may cover a set of base station cells, also referred to as base station sectors. That is, each cell may cover a sector (e.g., a 120° sector, etc.). Base station **120** include a radio frequency (RF) transceiver configured to send and receive wireless signals in the direction of the sector and be configured to communicate with UE devices **110** using a 5G NR air interface, a 4G LTE air interface, and/or using another type of cellular air interface.

(23) Core network **140** may be managed by a provider of cellular wireless communication services and may manage communication sessions of subscribers connecting to core network **140** via RAN **130**. For example, core network **140** may establish an Internet Protocol (IP) connection between UE devices **110** and PDN **150**. In some implementations, core network **140** may include a 5G core network. In other implementations, core network **140** may include a 4G core network (e.g., an evolved packet core (EPC) network).

(24) The components of core network **140** may be implemented as dedicated hardware components or as virtualized functions implemented on top of a common shared physical infrastructure using Software Defined Networking (SDN). For example, an SDN controller may implement one or more of the components of core network **140** using an adapter implementing a Virtual Network Function (VNF) virtual machine, a CNF container, an event driven serverless architecture interface, and/or another type of SDN component. The common shared physical infrastructure may be implemented using one or more devices **300** described below with reference to FIG. 3 in a cloud computing center associated with core network **140**. Exemplary components of core network **140** are described below with reference to FIG. 2.

(25) PDNs **150-A** to **150-Y** may each include a PDN connected to core network **140**. A particular PDN **150** may be associated with a DNN in 5G, and/or an APN in 4G, and a UE device may request a connection to PDN **150** using the DNN or APN. PDN **150** may include, and/or be connected to and enable communication with, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), an autonomous system (AS) on the Internet, an optical network, a cable television network, a satellite network, another wireless network (e.g., a CDMA network, a general packet radio service (GPRS) network, and/or an LTE network), an ad hoc network, a telephone network (e.g., the Public Switched Telephone Network (PSTN) or a cellular network), an intranet, or a combination of networks. PDN **150** may include an application server **160** (shown in PDN **150-A** in FIG. 1). Application server **160** may provide services for an application running on UE device **110** and may establish an application session with UE device **110** via an SME in core network **140**.

(26) Although FIG. 1 shows exemplary components of environment **100**, in other implementations, environment **100** may include fewer components, different components, differently arranged

components, or additional components than depicted in FIG. 1. Additionally, or alternatively, one or more components of environment **100** may perform functions described as being performed by one or more other components of environment **100**.

(27) FIG. 2 illustrates a system **200** that includes exemplary components of core network **140** in the context of environment **100** according to an implementation described herein in which core network **140** includes a 5G core network. As shown in FIG. 2, system **200** may include UE device **110**, gNodeB **210**, core network **140**, and PDN **150**.

(28) gNodeB **210** (corresponding to base station **120**) may include devices and components that enable UE device **110** to connect to core network **140** via RAN **130** using 5G NR RAT. For example, gNodeB **210** may service one or more cells, with each cell being served by a wireless transceiver with an antenna array configured for mm-wave wireless communication. gNodeB **210** may communicate with AMF **220** using an N2 interface **212** and communicate with UPF **230** using an N3 interface **214**.

(29) Core network **140** may include an Access and Mobility Function (AMF) **220**, a User Plane Function (UPF) **230**, an SMF **240**, a UDSF **245**, an Application Function (AF) **250**, a Unified Data Management (UDM) **252**, a Policy Control Function (PCF) **254**, a Charging Function (CHF) **256**, a Network Repository Function (NRF) **258**, a Network Exposure Function (NEF) **260**, a Network Slice Selection Function (NSSF) **262**, an Authentication Server Function (AUSF) **264**, a 5G Equipment Identity Register (EIR) **266**, a Network Data Analytics Function (NWDAF) **268**, a Short Message Service Function (SMSF) **270**, a Security Edge Protection Proxy (SEPP) **272**, and a Non-3GPP Inter-Working Function (N3IWF) **274**.

(30) While FIG. 2 depicts a single AMF **220**, UPF **230**, SMF **240**, AF **250**, UDM **252**, PCF **254**, CHF **256**, NRF **258**, NEF **260**, NSSF **262**, AUSF **264**, EIR **266**, NWDAF **268**, SMSF **270**, SEPP **272**, and N3IWF **274** for illustration purposes, in practice, core network **140** may include multiple AMFs **220**, UPFs **230**, SMFs **240**, AFs **250**, UDMs **252**, PCFs **254**, CHFs **256**, NRFs **258**, NEFs **260**, NSSFs **262**, AUSFs **264**, EIRs **266**, NWDAFs **268**, SMSFs **270**, SEPPs **272**, and/or N3IWFs **274**.

(31) AMF **220** may perform registration management, connection management, reachability management, mobility management, lawful intercepts, Short Message Service (SMS) transport between UE device **110** and SMSF **270**, transport of session management messages between UE device **110** and SMF **240**, access authentication and authorization, location services management, functionality to support non-3GPP access networks, and/or other types of management processes. AMF **220** may be accessible by other function nodes via an Namf interface **222**.

(32) UPF **230** may maintain an anchor point for intra/inter-RAT mobility, maintain an external Packet Data Unit (PDU) point of interconnect to a particular data network (e.g., PDN **150**), perform packet routing and forwarding, perform the user plane part of policy rule enforcement, perform packet inspection, perform lawful intercept, perform traffic usage reporting, perform QoS handling in the user plane, perform uplink traffic verification, perform transport level packet marking, perform downlink packet buffering, and/or perform other types of user plane processes. UPF **230** may communicate with SMF **240** using an N4 interface **232** and connect to PDN **160** using an N6 interface **234**.

(33) SMF **240** may perform session establishment, session modification, and/or session release, perform IP address allocation and management, perform Dynamic Host Configuration Protocol (DHCP) functions, perform selection and control of UPF **230**, configure traffic steering at UPF **230** to guide the traffic to the correct destinations, terminate interfaces toward PCF **254**, perform lawful intercepts, charge data collection, support charging interfaces, control and coordinate charging data collection, terminate session management parts of Non-Access Stratum (NAS) messages, perform downlink data notification, manage roaming functionality, and/or perform other types of control plane processes for managing user plane data. SMF **240** may be accessible via an Nsmf interface **242**.

(34) AF **250** may provide services associated with a particular application, such as, for example, an application for influencing traffic routing, an application for accessing NEF **260**, an application for interacting with a policy framework for policy control, and/or other types of applications. AF **250** may be accessible via an Naf interface **251**, also referred to as an NG5 interface.

(35) UDM **252** may maintain subscription information for UE devices **110**, manage subscriptions, generate authentication credentials, handle user identification, perform access authorization based on subscription data, perform network function registration management, maintain service and/or session continuity by maintaining assignment of SMF **240** for ongoing sessions, support SMS delivery, support lawful intercept functionality, and/or perform other processes associated with managing user data. UDM **252** may be accessible via a Nudm interface **253**.

(36) PCF **254** may support policies to control network behavior, provide policy rules to control plane functions (e.g., to SMF **240**), access subscription information relevant to policy decisions, perform policy decisions, and/or perform other types of processes associated with policy enforcement. PCF **254** may be accessible via Npcf interface **255**. PCF **254** may activate QoS monitoring for a communication session based on a trigger condition and instruct SMF **240** to perform QoS monitoring for the communication session. PCF **254** may collect QoS monitoring reports from SMFs **240** and send batch reports of the collected QoS monitoring reports to AF **250** and/or NEF **260**. CHF **256** may perform charging and/or billing functions for core network **140**. For example, CHF **256** may generate a charging record for UE device **110** based on data flow information associated with UE device **110**. CHF **256** may be accessible via Nchf interface **257**.

(37) NRF **258** may support a service discovery function and maintain profiles of available network function (NF) instances and their supported services. An NF profile may include, for example, an NF instance identifier (ID), an NF type, a Public Land Mobile Network (PLMN) ID associated with the NF, network slice IDs associated with the NF, capacity information for the NF, service authorization information for the NF, information identifying supported services associated with the NF, endpoint information for each supported service associated with the NF, and/or other types of NF information. NRF **258** may be accessible via an Nnrf interface **259**.

(38) NEF **260** may expose capabilities and events to other NFs, including 3.sup.rd party NFs, AFs, edge computing NFs, and/or other types of NFs. Furthermore, NEF **260** may secure provisioning of information from external applications to core network **140**, translate information between core network **140** and devices/networks external to core network **140**, support a Packet Flow Description (PFD) function, and/or perform other types of network exposure functions. NEF **260** may be accessible via Nnef interface **261**.

(39) NSSF **262** may select a set of network slice instances to serve a particular UE device **110**, determine network slice selection assistance information (NSSAI) or a Single-NSSAI (S-NSSA), determine a particular AMF **220** to serve a particular UE device **110**, and/or perform other types of processing associated with network slice selection or management. NSSF **262** may be accessible via Nnssf interface **263**.

(40) AUSF **264** may perform authentication. For example, AUSF **264** may implement an Extensible Authentication Protocol (EAP) authentication server and may store authentication keys for UE devices **110**. AUSF **264** may be accessible via Nausf interface **265**. EIR **266** may authenticate a particular UE device **110** based on UE device identity, such as a Permanent Equipment Identifier (PEI). For example, EIR **266** may check to determine if a PEI has been blacklisted. EIR **266** may be accessible via Neir interface **267**.

(41) NWDAF **268** may collect analytics information associated with radio access network **120** and/or core network **140**. For example, NWDAF **268** may collect accessibility Key Performance Indicators (KPIs) (e.g., a Radio Resource Control (RRC) connection setup success rate, a Radio Access Bearer (RAB) success rate, etc.), retainability KPIs (e.g., a call drop rate, etc.), mobility KPIs (e.g., a handover success rate, etc.), service integrity KPIs (e.g., downlink average throughput, downlink maximum throughput, uplink average throughput, uplink maximum

throughput, etc.), utilization KPIs (e.g., resource block utilization rate, average processor load, etc.), availability KPIs (e.g., radio network unavailability rate, etc.), traffic KPIs (e.g., downlink traffic volume, uplink traffic volume, average number of users, maximum number of users, a number of voice bearers, a number of video bearers, etc.), response time KPIs (e.g., latency, packet arrival time, etc.), and/or other types of wireless network KPIs.

(42) SMSF **270** may perform SMS services for UE devices **110**. SMSF **270** may be accessible via Nsmsf interface **271**. SEPP **272** may implement application layer security for all layer information exchanged between two NFs across two different PLMNs. N3IWF **274** may interconnect to a non-3GPP access device, such as, for example, a WiFi Access Point. N3IWF **274** may facilitate handovers for UE device **110** between RAN **130** and the non-3GPP access device. N3IWF **274** may be accessible via Nn3iwf interface **275**.

(43) Although FIG. **2** shows exemplary components of core network **140**, in other implementations, core network **140** may include fewer components, different components, differently arranged components, or additional components than depicted in FIG. **2**. Additionally, or alternatively, one or more components of core network **140** may perform functions described as being performed by one or more other components of core network **140**. For example, core network **140** may include additional function nodes not shown in FIG. **2**, such as a Unified Data Repository (UDR), an Unstructured Data Storage Network Function (UDSF), a Location Management Function (LMF), a Lawful Intercept Function (LIF), a Binding Session Function (BSF), and/or other types of functions. Furthermore, while particular interfaces have been described with respect to particular function nodes in FIG. **2**, additionally, or alternatively, core network **140** may include a reference point architecture that includes point-to-point interfaces between particular function nodes.

(44) FIG. **3** illustrates example components of a device **300** according to an implementation described herein. UE device **110**, application server **160**, gNodeB **210**, AMF **220**, UPF **230**, SMF **240**, AF **250**, UDM **252**, PCF **254**, CHF **256**, NRF **258**, NEF **260**, NSSF **262**, AUSF **264**, EIR **266**, NWDAF **268**, SMSF **270**, SEPP **272**, N3IWF **274**, and/or other components of core network **140**, may each include one or more devices **300**. As shown in FIG. **3**, device **300** may include a bus **310**, a processor **320**, a memory **330**, an input device **340**, an output device **350**, and a communication interface **360**.

(45) Bus **310** may include a path that permits communication among the components of device **300**. Processor **320** may include any type of single-core processor, multi-core processor, microprocessor, latch-based processor, and/or processing logic (or families of processors, microprocessors, and/or processing logics) that interprets and executes instructions. In other embodiments, processor **320** may include an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another type of integrated circuit or processing logic.

(46) Memory **330** may include any type of dynamic storage device that may store information and/or instructions, for execution by processor **320**, and/or any type of non-volatile storage device that may store information for use by processor **320**. For example, memory **330** may include a random access memory (RAM) or another type of dynamic storage device, a read-only memory (ROM) device or another type of static storage device, a content addressable memory (CAM), a magnetic and/or optical recording memory device and its corresponding drive (e.g., a hard disk drive, optical drive, etc.), and/or a removable form of memory, such as a flash memory.

(47) Input device **340** may allow an operator to input information into device **300**. Input device **340** may include, for example, a keyboard, a mouse, a pen, a microphone, a remote control, an audio capture device, an image and/or video capture device, a touch-screen display, and/or another type of input device. In some embodiments, device **300** may be managed remotely and may not include input device **340**. In other words, device **300** may be “headless” and may not include a keyboard, for example.

(48) Output device **350** may output information to an operator of device **300**. Output device **350** may include a display, a printer, a speaker, and/or another type of output device. For example,

device **300** may include a display, which may include a liquid-crystal display (LCD) for displaying content to the customer. In some embodiments, device **300** may be managed remotely and may not include output device **350**. In other words, device **300** may be “headless” and may not include a display, for example.

(49) Communication interface **360** may include a transceiver that enables device **300** to communicate with other devices and/or systems via wireless communications (e.g., radio frequency, infrared, and/or visual optics, etc.), wired communications (e.g., conductive wire, twisted pair cable, coaxial cable, transmission line, fiber optic cable, and/or waveguide, etc.), or a combination of wireless and wired communications. Communication interface **360** may include a transmitter that converts baseband signals to RF signals and/or a receiver that converts RF signals to baseband signals. Communication interface **360** may be coupled to one or more antennas/antenna arrays for transmitting and receiving RF signals.

(50) Communication interface **360** may include a logical component that includes input and/or output ports, input and/or output systems, and/or other input and output components that facilitate the transmission of data to other devices. For example, communication interface **360** may include a network interface card (e.g., Ethernet card) for wired communications and/or a wireless network interface (e.g., a WiFi) card for wireless communications. Communication interface **360** may also include a universal serial bus (USB) port for communications over a cable, a Bluetooth™ wireless interface, a radio-frequency identification (RFID) interface, a near-field communications (NFC) wireless interface, and/or any other type of interface that converts data from one form to another form.

(51) As will be described in detail below, device **300** may perform certain operations relating to activating QoS monitoring and/or batch reporting of QoS monitoring reports. Device **300** may perform these operations in response to processor **320** executing software instructions contained in a computer-readable medium, such as memory **330**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may be implemented within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **330** from another computer-readable medium or from another device. The software instructions contained in memory **330** may cause processor **320** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of, or in combination with, software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(52) Although FIG. 3 shows exemplary components of device **300**, in other implementations, device **300** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. 3. Additionally, or alternatively, one or more components of device **300** may perform one or more tasks described as being performed by one or more other components of device **300**.

(53) FIG. 4 is a diagram illustrating exemplary components of PCF **254**. The components of PCF **254** may be implemented, for example, via processor **320** executing instructions from memory **330**. Alternatively, some or all of the components of PCF **254** may be implemented via hard-wired circuitry. As shown in FIG. 4, PCF **254** may include a UDM interface **410**, a CHF interface **412**, an SMF interface **414**, an AF/NEF interface **416**, a QoS monitoring manager **420**, a QoS monitoring criteria database (DB) **425**, a QoS policy manager **430**, a QoS policy rules DB **435**, a QoS monitoring reports DB **445**, a batch reporting manager **450**, and a batch reporting rules DB **455**.

(54) UDM interface **410** may be configured to communicate with UDM **252**. For example, UDM interface **410** may receive a subscriber update from UDM **252** indicating that QoS monitoring is to be used for communication sessions associated with a subscriber. CHF interface **412** may be configured to communicate with CHF **256**. For example, CHF interface **412** may receive a request from CHF **256** to activate QoS monitoring for a particular communication session.

(55) SMF interface **414** may be configured to communicate with SMF **240**. For example, SMF

interface **414** may receive a request from SMF **240** for a policy for a communication session, send a QoS monitoring policy to SMF **240**, and/or receive a QoS monitoring report based on the QoS monitoring policy and store the received QoS monitoring report in QoS monitoring reports DB **445**. AF/NEF interface **416** may be configured to communicate with AF **250** and/or NEF **260**. For example, AF/NEF interface **416** may receive a request to perform QoS monitoring for a communication session from AF **250** and/or NEF **260**. Furthermore, AF/NEF interface **416** may send a batch report that includes information from multiple QoS monitoring reports to AF **250** and/or NEF **260**.

(56) QoS monitoring manager **420** may manage QoS monitoring for communication sessions for which PCF **254** handles policies. As an example, QoS monitoring manager **420** may receive a policy authorization request from AF **250** or from NEF **260**. The request may instruct PCF **254** to monitor a communication session for one or more QoS parameters. As another example, QoS monitoring manager **420** may determine to activate QoS monitoring for a communication session based on a trigger criterion. QoS monitoring criteria DB **425** may store information relating to trigger criteria for initiating QoS monitoring for a communication session. Exemplary information that may be stored in QoS monitoring criteria DB **425** is described below with reference to FIG. 5A. QoS monitoring manager **420** may send a QoS monitoring request to SMF **240** based the trigger criterion. The QoS monitoring request may identify one or more QoS parameters, for which SMF **240** is to collect values for the communication session, and identify a frequency at which to collect the values.

(57) QoS policy manager **430** may manage QoS policies for communication sessions. For example, QoS policy manager **430** may select an action based on a received QoS monitoring report and perform the selected action. QoS policy rules DB **435** may store information relating to actions that are to be taken based on information received in a QoS monitoring report. Exemplary information that may be stored in QoS policy rules DB **435** is described below with reference to FIG. 5B. The action to be taken may include reporting the QoS monitoring report to AF **250** or NEF **260**, adding the QoS monitoring report to a batch report to be sent to AF **250** or NEF **260** at a later time, and/or applying a policy to the communication session, such as changing a QoS class for the communication session, instructing the SMF to collect additional QoS information relating to the communication session, instructing the SMF to transfer the communication session to a particular UPF, assigning the communication session to a particular network slice, etc.

(58) QoS monitoring reports DB **445** may store information relating to QoS monitoring reports received from SMF **240**. For example, for each QoS monitoring report received by PCF **254**, QoS monitoring reports DB **445** may store a record that includes information identifying a communication session associated with the QoS monitoring report, information identifying UE device **110** associated with the communication session, information identifying a QoS class associated with the communication session, information identifying one or more parameters associated with the communication session, and/or information identifying values for one or more QoS parameters collected by UPF **230** and provided to SMF **240** to include in the QoS monitoring report. The QoS parameters may include uplink delay, downlink delay, round-trip delay, packet loss rate, jitter rate, throughput rate, and/or another QoS parameter associated with the communication session.

(59) Batch reporting manager **450** may manage batch reporting of QoS monitoring reports based on one or more batch reporting rules stored in batch reporting rules DB **455**. A batch reporting rule may specify a reporting trigger condition for sending a batch report to AF **250** or to NEF **260**. A trigger condition may include a data size threshold. For example, when the size of accumulated data included in received QoS monitoring reports (e.g., received since a most recent batch report was sent) reaches the data size threshold, batch reporting manager **450** may generate a batch report that includes the received QoS monitoring reports and send the batch report to AF **250** or NEF **260**. Another trigger condition may include a threshold number of QoS monitoring reports. For

example, when PCF **254** receives the threshold number of QoS monitoring reports since a most recent batch report was sent, batch reporting manager **450** may generate a batch report that includes all QoS monitoring reports received since the most recent batch report was sent and send the batch report to AF **250** or NEF **260**. Yet another trigger condition may include detecting that a particular time period has elapsed. For example, when a time period since a most recent batch report was sent has elapsed, batch reporting manager **450** may generate a batch report that includes all QoS monitoring reports received since the most recent batch report was sent and send the batch report to AF **250** or NEF **260**. A trigger condition may be applied to all QoS monitoring reports received by PCF **254** or to QoS monitoring reports received for a particular subscriber (e.g., a particular UE device **110**).

(60) Although FIG. **4** shows exemplary components of PCF **254**, in other implementations, PCF **254** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. **4**. Additionally, or alternatively, one or more components of PCF **254** may perform one or more tasks described as being performed by one or more other components of PCF **254**.

(61) FIG. **5A** is a diagram illustrating exemplary components of QoS monitoring criteria DB **425**. As shown in FIG. **5A**, QoS monitoring criteria DB **425** may include trigger condition records **500**. Each trigger condition record **500** may store information relating to a particular trigger condition for triggering QoS monitoring. Trigger condition record **500** may include a triggering criterion identifier (ID) field **510**, a triggering criterion type field **520**, a QoS monitoring parameters field **530**, and a reporting frequency field **540**.

(62) Triggering criterion ID field **510** may store an ID associated with a particular triggering criterion. Triggering criterion type field **520** may identify one or more trigger criteria for initiating QoS monitoring for a communication session. As an example, a trigger criterion may include determining that a communication session is associated with a particular subscriber, UE device **110**, a QoS class, a network slice, a DNN, gNodeB **210** and/or another parameter or device in RAN **130**, core network **140**, and/or PDN **150**. As another example, a trigger condition may further include receiving, from a CHF, a notification from indicating that QoS monitoring is to be performed for the communication session, receiving a subscriber profile update indicating that QoS monitoring is to be used for communication sessions associated with a subscriber, and/or receiving another notification from a component of core network **140**.

(63) QoS monitoring parameters field **530** may identify one or more QoS monitoring parameters that are to be measured for the communication session if the communication session satisfies the particular trigger condition. For example, QoS monitoring parameters field **530** may identify one or more of an uplink delay, a downlink delay, a round-trip delay, a packet loss rate, a jitter rate, a throughput rate, and/or another QoS parameter. Reporting frequency field **540** may store information identifying how frequently to send a QoS monitoring report for the communication session. For example, reporting frequency field **540** may specify that a QoS monitoring report is to be sent once per communication session, at periodic intervals while the communication session is active, as an average of multiple measurements taken while the communication session is active and sent once the communication session is released, and/or another type of reporting frequency.

(64) Although FIG. **5A** shows exemplary components of QoS monitoring criteria DB **425**, in other implementations, QoS monitoring criteria DB **425** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. **5A**.

(65) FIG. **5B** is a diagram illustrating exemplary components of QoS policy rules DB **435**. As shown in FIG. **5B**, QoS policy rules DB **435** may include action records **550**. Each action record **550** may store information relating to a particular action to take in response to information included in a received QoS monitoring report. Action record **550** may include a QoS report action trigger ID field **560**, an action trigger criteria field **570**, and an action field **580**.

(66) QoS report action trigger ID field **560** may include an ID associated with a particular action

trigger. Action trigger criteria field **570** may store information identifying one or more criteria for the particular action trigger. The criteria may include, for example, a threshold or range of values for a particular QoS parameter for a type of communication session. Action field **580** may identify actions to be performed if the criteria included in action trigger criteria field **750** are satisfied. The action may include, for example, the actions described above with respect to QoS policy manager **430**.

(67) Although FIG. 5B shows exemplary components of QoS policy rules DB **435**, in other implementations, QoS policy rules DB **435** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. 5B.

(68) FIG. 6 illustrates a flowchart for initiating QoS monitoring according to an implementation described herein. In some implementations, process **600** of FIG. 6 may be performed by PCF **254**. In other implementations, some or all of process **600** may be performed by another device or a group of devices.

(69) As shown in FIG. 6, process **600** may include receiving a request from an SMF for a policy for a session (block **610**) and determining that QoS monitoring is to be activated for the session (block **620**). For example, SMF **240** may, during the establishment of a communication session, send a policy request to PCF **254** for a policy to apply to the communication session. In response, PCF **254** may determine that QoS monitoring is to be activated for the communication session by determining that a QoS monitoring criterion in QoS monitoring criteria DB **425** is satisfied. As an example, PCF **254** may determine that the communication session is associated with a subscriber for whom QoS monitoring is to be performed. As another example, PCF **254** may determine that the communication session is associated with a DNN for which QoS monitoring is to be performed.

(70) Process **600** may further include providing a QoS monitoring policy to the SMF (block **630**), receiving a QoS monitoring report from the SMF (block **640**), and performing an action based on the received QoS monitoring report (block **650**). For example, PCF **254** may send a QoS monitoring request to SMF **240**, receive a QoS monitoring report in response, and perform an action based on information stored in QoS policy rules DB **435**. For example, PCF **254** may send the QoS monitoring report to AF **250** or NEF **260**, add the QoS monitoring report to a batch report to be sent to AF **250** or NEF **260** at a later time, and/or apply a policy to the communication session, such as changing a QoS class for the communication session, instructing the SMF to collect additional QoS information relating to the communication session, instructing the SMF to transfer the communication session to a particular UPF, assigning the communication session to a particular network slice, etc.

(71) FIG. 7 illustrates a flowchart for batch reporting of QoS monitoring reports according to an implementation described herein. In some implementations, process **700** of FIG. 7 may be performed by PCF **254**. In other implementations, some or all of process **700** may be performed by another device or a group of devices.

(72) As shown in FIG. 7, process **700** may include generating a batch reporting rule (block **710**). For example, an administrator may define a batch reporting rule and store the batch reporting rule in batch reporting rules DB **455**. The batch reporting rule may specify a trigger for generating a batch report of multiple QoS monitoring reports to send to AF **250** or NEF **260**. The trigger may include, for example, a data size threshold for total data accumulated in received QoS monitoring reports, a threshold number of received QoS monitoring reports, a passage of a particular length of time since a previous batch report was sent, and/or another type of trigger condition that applies to all received QoS monitoring reports or to QoS monitoring reports received for a particular subscriber or UE device **110**.

(73) Process **700** may further include providing a QoS monitoring policy to an SMF (block **720**) and collecting QoS monitoring reports from the SMF (block **730**). For example, PCF **254** may provide a QoS monitoring policy to SMFs **240** for communication sessions. PCF **254** may collect QoS monitoring reports from the SMF s **240** for the communication sessions over a period of time.

(74) Process **700** may further include detecting that a reporting trigger condition for the batch reporting rule is satisfied (block **740**) and generating and sending a batch report in response (block **750**). As an example, PCF **254** may determine that a particular number of QoS monitoring reports have been received for a particular UE device **110** and may generate a batch report that includes QoS parameter values from all the QoS monitoring reports and send the batch report to NEF **260**. NEF **260** may forward the batch report to application server **160**. As another example, PCF **254** may determine that a particular time period has passed since the last batch report was sent to AF **250** for UE device **110**. In response, PCF **254** may generate a batch report that includes QoS parameter values from all the QoS monitoring reports received since the last batch report and send the batch report to AF **250**.

(75) FIG. **8** illustrates an exemplary signal flow **800** according to an implementation described herein. Signal flow **800** illustrates QoS monitoring initiated by PCF **254**. As shown in FIG. **8**, signal flow **800** may include SMF **240** sending a policy request to PCF **254** for a communication session managed by SMF **240** (signal **810**). In response, PCF **254** may activate QoS monitoring for the communication session based on identifying UE device **110** is associated with the communication session and determining QoS monitoring is to be activated for communication session associated with UE device **110**. PCF **254** may send a QoS monitoring policy that specifies one or more QoS parameters to monitor for the communication session (signal **830**). SMF **240** may enable QoS monitoring with UPF **230** managing the communication session (signal **832**). UPF **230** may measure values for the one or more QoS parameters, such as uplink delay, downlink delay, roundtrip delay, etc., for the communication session and provide the measured values to SMF **240** (signal **834**). SMF **240** may generate a QoS monitoring report and send the QoS monitoring report to PCF **254** (signal **840**).

(76) Assume that the QoS monitoring report includes a latency value that is larger than a latency requirement associated with the communication session. Therefore, the communication session may not be meeting the latency requirement. Thus, based on the received QoS monitoring report, PCF **254** may select an action to adjust the QoS class for the communication session to a higher QoS class (block **850**) and send an updated policy to SMF **240** (signal **852**) and SMF **240** may apply the updated policy to UPF **230** (signal **854**).

(77) While FIG. **8** illustrates a situation in which PCF **254** send a QoS monitoring request to SMF **240** when a communication session is established, in other situations PCF **254** may send a QoS monitoring request/policy to SMF **240** after a communication session has been established and is running (e.g., in mid-session, etc.). For example, PCF **254** may receive a subscriber profile change from UDR via UDM **252** while in mid-session, and the subscriber profile change may indicate that QoS monitoring is to be used for the subscriber for communication sessions of a particular type. In response, PCF **254** may send the QoS monitoring policy to SMF **240** while the communication session is running.

(78) FIG. **9** illustrates another first exemplary signal flow **900** according to an implementation described herein. Signal flow **900** illustrates batch reporting of QoS monitoring reports. As shown in FIG. **9**, signal flow **900** may include AF **250** or NEF **260** sending a policy authorization request to PCF **254** to initiate QoS monitoring for a communication session (signal **910**). PCF **254** may respond with a policy authorization response acknowledging receipt of the request (signal **920**).

(79) PCF **254** may then, in response, collect a QoS monitoring report (item **930-A**). Collecting a QoS monitoring report may include PCF **254** providing a QoS monitoring policy to SMF **240** (signal **932-A**), SMF **240** enabling QoS monitoring with UPF **230** handling the communication session (signal **934-A**), receiving a QoS monitoring report from UPF **230** (signal **936-A**), forwarding the QoS monitoring report to PCF **254** (signal **938-A**), and collecting and storing the QoS monitoring report (block **940-A**).

(80) PCF **254** may collect and store multiple QoS monitoring reports until batch reporting is triggered (block **950**). In response, PCF **254** may generate a batch report that includes the QoS

parameter values from the collected QoS monitoring reports (block **960**) and send the batch report to AF **250** or NEF **260** (signal **970**).

(81) In the preceding specification, various preferred embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(82) For example, while a series of blocks have been described with respect to FIGS. **6** and **7**, and a series of signals have been described with respect to FIGS. **8** and **9**, the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel.

(83) It will be apparent that systems and/or methods, as described above, may be implemented in many different forms of software, firmware, and hardware in the implementations illustrated in the figures. The actual software code or specialized control hardware used to implement these systems and methods is not limiting of the embodiments. Thus, the operation and behavior of the systems and methods were described without reference to the specific software code—it being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

(84) Further, certain portions, described above, may be implemented as a component that performs one or more functions. A component, as used herein, may include hardware, such as a processor, an ASIC, or a FPGA, or a combination of hardware and software (e.g., a processor executing software).

(85) It should be emphasized that the terms “comprises”/“comprising” when used in this specification are taken to specify the presence of stated features, integers, steps or components but does not preclude the presence or addition of one or more other features, integers, steps, components or groups thereof.

(86) The term “logic,” as used herein, may refer to a combination of one or more processors configured to execute instructions stored in one or more memory devices, may refer to hardwired circuitry, and/or may refer to a combination thereof. Furthermore, a logic may be included in a single device or may be distributed across multiple, and possibly remote, devices.

(87) For the purposes of describing and defining the present invention, it is additionally noted that the term “substantially” is utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. The term “substantially” is also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

(88) To the extent the aforementioned embodiments collect, store, or employ personal information of individuals, it should be understood that such information shall be collected, stored, and used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

(89) No element, act, or instruction used in the present application should be construed as critical or essential to the embodiments unless explicitly described as such. Also, as used herein, the article “a” is intended to include one or more items. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A method comprising: determining, by a device, that Quality of Service (QoS) monitoring is to be performed for a communication session; generating, by the device, a QoS monitoring policy for the communication session; providing, by the device, the QoS monitoring policy to a Session Management Function (SMF) associated with the communication session; receiving, by the device, a QoS monitoring report for the communication session from the SMF; and performing, by the device, an action based on the received QoS monitoring report, wherein determining that QoS monitoring is to be performed for the communication session includes at least one of: receiving, from a Charging Function (CHF), a notification indicating that QoS monitoring is to be performed for the communication session; receiving, from a Unified Data Management (UDM), a subscriber profile update indicating that a subscriber associated with the communication session is to receive QoS monitoring; or determining that QoS monitoring is to be established for communication sessions associated with a particular Data Network Name (DNN).
2. The method of claim 1, wherein the device includes a Policy Control Function (PCF).
3. The method of claim 1, wherein the QoS monitoring policy instructs the SMF to measure at least one of: an uplink delay, a downlink delay, a round-trip delay, a packet loss rate, a jitter rate, or a throughput rate.
4. The method of claim 1, wherein performing the action based on the received QoS monitoring report includes at least one of: sending the received QoS monitoring report to an Application Function (AF) or a Network Exposure Function (NEF); or applying a policy to the SMF based on the received QoS monitoring report.
5. The method of claim 4, wherein applying the policy to the SMF based on the received QoS monitoring report includes at least one of: changing a QoS class for the communication session; instructing the SMF to collect additional QoS information relating to the communication session; instructing the SMF to transfer the communication session to a particular User Plane Function (UPF); or assigning the communication session to a particular network slice.
6. The method of claim 1, further comprising: collecting a plurality of QoS monitoring reports from one or more SMFs; and reporting the plurality of QoS monitoring reports to an Application Function (AF) or a Network Exposure Function (NEF) in a batch report.
7. The method of claim 6, wherein reporting the plurality of QoS monitoring reports to the AF or NEF in a batch report includes: generating a batch reporting rule, wherein the batch reporting rule includes a reporting trigger condition; detecting that the reporting trigger condition is satisfied; and sending the batch report to the AF or NEF, in response to detecting that the reporting trigger condition is satisfied.
8. The method of claim 7, wherein detecting that the reporting trigger condition is satisfied includes at least one of: detecting that a size of data included in the plurality of QoS monitoring reports has reached a data size threshold; detecting that a particular time period has elapsed; or detecting that a particular number of QoS monitoring reports have been received for a particular subscriber.
9. A device comprising: a processor configured to: determine that Quality of Service (QoS) monitoring is to be performed for a communication session; generate a QoS monitoring policy for the communication session; provide the QoS monitoring policy to a Session Management Function (SMF) associated with the communication session; receive a QoS monitoring report for the communication session from the SMF; and perform an action based on the received QoS monitoring report, wherein when determining that QoS monitoring is to be performed for the communication session, the processor is further configured to at least one of: receive, from a Charging Function (CHF), a notification indicating that QoS monitoring is to be performed for the communication session; receive, from a Unified Data Management (UDM), a subscriber profile update indicating that a subscriber associated with the communication session is to receive QoS

monitoring; or determine that QoS monitoring is to be established for communication sessions associated with a particular Data Network Name (DNN).

10. The device of claim 9, wherein the device includes a Policy Control Function (PCF).

11. The device of **9**, wherein the QoS monitoring policy instructs the SMF to measure at least one of: an uplink delay, a downlink delay, a round-trip delay, a packet loss rate, a jitter rate, or a throughput rate.

12. The device of claim 9, wherein, when performing the action based on the received QoS monitoring report, the processor is further configured to at least one of: send the received QoS monitoring report to an Application Function (AF) or a Network Exposure Function (NEF); or apply a policy to the SMF based on the received QoS monitoring report.

13. The device of claim 12, wherein, when applying the policy to the SMF based on the received QoS monitoring report, the processor is further configured to at least one of: change a QoS class for the communication session; instruct the SMF to collect additional QoS information relating to the communication session; instruct the SMF to transfer the communication session to a particular User Plane Function (UPF); or assign the communication session to a particular network slice.

14. The device of claim 9, wherein the processor is further configured to: collect a plurality of QoS monitoring reports from one or more SMFs; and report the plurality of QoS monitoring reports to an Application Function (AF) or a Network Exposure Function (NEF) in a batch report.

15. The device of claim 14, wherein, when reporting the plurality of QoS monitoring reports to the AF or NEF in a batch report, the processor is further configured to: generate a batch reporting rule, wherein the batch reporting rule includes a reporting trigger condition; detect that the reporting trigger condition is satisfied; and send the batch report to the AF or NEF, in response to detecting that the reporting trigger condition is satisfied.

16. The device of claim 15, wherein, when detecting that the reporting trigger condition is satisfied, the processor is further configured to at least one of: detect that a size of data included in the plurality of QoS monitoring reports has reached a data size threshold; detect that a particular time period has elapsed; or detect that a particular number of QoS monitoring reports have been received for a particular subscriber.

17. A non-transitory computer-readable memory device storing instructions executable one a processor, the non-transitory computer-readable memory device comprising: one or more instructions to determine that Quality of Service (QoS) monitoring is to be performed for a communication session; one or more instructions to generate a QoS monitoring policy for the communication session; one or more instructions to provide the QoS monitoring policy to a Session Management Function (SMF) associated with the communication session; one or more instructions to receive a QoS monitoring report for the communication session from the SMF; and one or more instructions to perform an action based on the received QoS monitoring report, wherein the one or more instructions to determine that QoS monitoring is to be performed for the communication session comprise at least one of: one or more instructions to receive, from a Charging Function (CHF), a notification indicating that QoS monitoring is to be performed for the communication session; one or more instructions to receive, from a Unified Data Management (UDM), a subscriber profile update indicating that a subscriber associated with the communication session is to receive QoS monitoring; or one or more instructions to determine that QoS monitoring is to be established for communication sessions associated with a particular Data Network Name (DNN).

18. The non-transitory computer-readable memory device of claim 17, further comprising: one or more instructions to collect a plurality of QoS monitoring reports from one or more SMFs; and one or more instructions to report the plurality of QoS monitoring reports to an Application Function (AF) or a Network Exposure Function (NEF) in a batch report.

19. The non-transitory computer-readable memory device of claim 17, wherein the QoS monitoring policy instructs the SMF to measure at least one of: an uplink delay, a downlink delay, a round-trip

delay, a packet loss rate, a jitter rate, or a throughput rate.

20. The non-transitory computer-readable memory device of claim 17, wherein the one or more instructions to perform the action based on the received QoS monitoring report includes at least one of: one or more instructions to send the received QoS monitoring report to an Application Function (AF) or a Network Exposure Function (NEF); or one or more instructions to apply a policy to the SMF based on the received QoS monitoring report.
